AUTOSAR HIGH-LEVEL DESIGN (HLD) SPECIFICATION

System Component: Body Control Module (BCM) Software Architecture

Document Ref: BCM-HLD-AUTOSAR-REL-2026-v2.4 | Status: Approved Baseline | Safety Class: ISO 26262 ASIL B

1. System Overview and Scope

This High-Level Design document specifies the AUTOSAR Classic Release 4.4 software architecture for the Body Control Module (BCM). The BCM ECU oversees essential body electronics including exterior lighting, intelligent wiper control, central locking, and low-power vehicle sleep state transitions. The software architecture conforms strictly to AUTOSAR layered standards, partitioning functionality into Application Software Components (Application SWCs), Sensor-Actuator SWCs, and Basic Software (BSW) services bridged by the Virtual Functional Bus (VFB) and AUTOSAR Runtime Environment (RTE).

Target Hardware Platform: Infineon AURIX TC397 32-bit TriCore, dual-lockstep cores with HSM security extension.

2. Software Component (SWC) Architecture Inventory

The BCM application layer comprises four core software components:

Component Name (SWC)	SWC Type	Safety Level	Primary Functional Responsibility
`SWC_ExteriorLighting`	Application SWC	ASIL B	Controls Low Beam, High Beam, DRL, Turn Indicators, and Emergency Hazard Flasher logic.
`SWC_WiperControl`	Sensor-Actuator SWC	ASIL A	Interprets optical rain intensity, manages intermittent/continuous wiper sweep, and wash pump actuation.
`SWC_CentralLocking`	Application SWC	QM	Processes Remote Keyless Entry (RKE) RF signals, manages door lock motors, and crash unlock override.
`SWC_PowerManagement`	Service Proxy SWC	ASIL B	Monitors 12V terminal voltages (KL30/KL15), coordinates network sleep/wake triggers, and load shedding.

3. Detailed Port and Interface Specifications

Communication between SWCs is strictly conducted across the RTE via typed Sender-Receiver (S/R) and Client-Server (C/S) interfaces. The following matrix defines the approved ports, referenced interface prototypes, payload data types, and initial values.

Component	Port Prototype	Direction	Interface Name	Data Element	Data Type	Init Value
SWC_ExteriorLighting	R_AmbientLight	R-Port	If_AmbientLight	LightLevelLux	uint16	0
SWC_ExteriorLighting	R_VehicleSpeed	R-Port	If_VehicleSpeed	SpeedKmh	uint16	0
SWC_ExteriorLighting	P_LowBeamStatus	P-Port	If_LowBeamStatus	LowBeamActive	boolean	FALSE
SWC_ExteriorLighting	P_HighBeamStatus	P-Port	If_HighBeamStatus	HighBeamActive	boolean	FALSE
SWC_ExteriorLighting	R_HazardLightCmd	R-Port	If_HazardCommand	HazardActive	boolean	FALSE
SWC_WiperControl	R_RainIntensity	R-Port	If_RainIntensity	RainPercent	uint8	0
SWC_WiperControl	R_VehicleSpeed	R-Port	If_VehicleSpeed	SpeedKmh	uint16	0
SWC_WiperControl	P_WiperMotorCmd	P-Port	If_WiperMotorCmd	MotorSpeedState	uint8	0 (OFF)
SWC_WiperControl	P_WashPumpControl	P-Port	If_WashPumpControl	PumpActive	boolean	FALSE
SWC_CentralLocking	R_DoorAjarSensors	R-Port	If_DoorStatus	DoorAjarMask	uint8	0x00
SWC_CentralLocking	P_LockActuators	P-Port	If_LockActuator	LockCommand	uint8	0 (UNLOCKED)
SWC_CentralLocking	R_CrashSignal	R-Port	If_CrashEvent	CrashConfirmed	boolean	FALSE
SWC_PowerManagement	P_VehicleSpeed	P-Port	If_VehicleSpeed	SpeedKmh	uint8	0
SWC_PowerManagement	P_BatteryVoltage	P-Port	If_BatteryState	VoltageMilliVolts	uint16	12600
SWC_PowerManagement	P_ECUSleepState	P-Port	If_PowerState	SleepMode	uint8	1 (RUN)

4. End-to-End Functional Flows and Traceability

Flow 1: Rain-Activated Automatic Wiper Sweep (ASIL A Traceability)

1. Rain sensor optical photodiode measures infrared attenuation and publishes raw optical loss via LIN bus.
2. LIN BSW Driver passes signal to RTE, which provides data to SWC_WiperControl via port R_RainIntensity.
3. Runnable entity 'RE_WiperSpeedArbitration' executes every 20ms, evaluating current RainPercent against hysteresis threshold (Threshold_Low = 15%, Threshold_High = 60%).
4. Vehicle speed is simultaneously read via R_VehicleSpeed. If SpeedKmh > 80 km/h, wiper frequency is elevated to next stage.
5. Output command is published to port P_WiperMotorCmd (0=OFF, 1=Intermittent, 2=Low, 3=High) to drive H-Bridge FET actuators.

Flow 2: High-Beam Auto Switching Flow

1. Forward-facing camera evaluates oncoming glare and ambient illuminance, transmitted via CAN-FD to port R_AmbientLight.
2. SWC_ExteriorLighting runnable 'RE_LightingManager' triggers every 10ms. When LightLevelLux < 50 Lux and SpeedKmh > 40 km/h, P_HighBeamStatus is asserted TRUE.
3. When oncoming vehicle detected or vehicle speed drops below 30 km/h, P_HighBeamStatus transitions to FALSE with 250ms debouncing.

5. Runnable Entities and RTE Scheduling Configuration

Runnable Entity Name	Owning SWC	Activation Event	Period / Trigger	OS Task Assignment
`RE_LightingManager`	SWC_ExteriorLighting	TimingEvent	10 ms cyclic	Task_Periodic_10ms
`RE_WiperSpeedArbitration`	SWC_WiperControl	TimingEvent	20 ms cyclic	Task_Periodic_20ms
`RE_LockStateArbitration`	SWC_CentralLocking	DataReceivedEvent	On R_CrashSignal	Task_Event_HighPri
`RE_PowerStateMonitoring`	SWC_PowerManagement	TimingEvent	100 ms cyclic	Task_Periodic_100ms

6. Architectural Traceability and Known Design Constraints

Section 6.1 Known Inconsistencies & Open Architecture Review Items:

• **Discrepancy INC-01 (Data Type Mismatch):** SWC_ExteriorLighting and SWC_WiperControl declare port R_VehicleSpeed expecting interface If_VehicleSpeed with data element SpeedKmh of type **uint16**. However, SWC_PowerManagement exposes providing port P_VehicleSpeed with data element SpeedKmh of type **uint8**. This 8-bit to 16-bit truncation will cause overflow clipping if vehicle speed exceeds 255 km/h.

• **Discrepancy INC-02 (Dangling Required Port):** SWC_ExteriorLighting defines required port R_HazardLightCmd referencing If_HazardCommand. In the current BCM architecture v2.4, there is no corresponding Providing Port (P-Port) implemented on any active SWC, leading to an unresolved VFB connection.

• **Discrepancy INC-03 (Missing Signal Mapping):** SWC_CentralLocking requires R_CrashSignal from the Airbag Controller over CAN. The gateway routing table lacks a PDU mapping for CAN-ID 0x140 into the RTE virtual signal bus.

Document Version History:

• v2.4 (Current): Added SWC_WiperControl WashPump actuation, updated ASIL ratings.
• v2.3 (Legacy): Original BCM baseline with legacy SWC_AnalogLightingLegacy.
• v2.0: Initial concept release.